EPFL — Fall 2026

PHYS-344

Quantum mechanics for non-physicists

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Lecture notes — Monday 7 September 2026

This document was originally created for my own use, but I am sharing it because I believe it may be useful to others for studying and reference purposes. However mistake can happen and there will inevitably be some errors, if you found any, you are more than welcome to send me a message (Arthur Hbe on telegram) or to directly do a PR on the repo

`https://github.com/Arthur926564/LectureNotes`

If you think that something is missing or that something is not clear, I will be more than happy to add your modification to this document.

Unless otherwise stated, all material in these notes is based on the course content, including the diagrams, explanations, and examples.

Note also that these notes were taken during the lecture, but not only; some materials was added later (that's why you may read remarks that sounds like I knew the course before taking it).

Finally, thanks a lot to Joachim Favre for his help with the setup and whose style is used in this document.

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Introduction

Overview of the course

Main textbook

Chapter 1

Introduction

1.1 The linear algebra of qbits

Math of 2-level systems (Qbits)

Two describe our system, we will need a vector space of two elements (x_0, z_0). Furthermore, $x_0, z_0 \in \mathbb{C}$. In regular mathematical notation, we would just call our vector $\vec{s}$. On the other hand, quantum mechanics uses Dirac notation, we will call our vector:

$$|\varphi\rangle$$

Ket

To do so we will also need to define **basis** vectors, which we will call $|0\rangle, |1\rangle$. We can easily define them as

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

What this mean is that for any vector $|\psi\rangle$ we can always define it as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Bra

Definition 1. Bra Given a vector $|\varphi\rangle = \alpha|0\rangle + \beta|1\rangle$ we can define its **bra** as:

$$\langle\varphi| = |\varphi\rangle^\dagger = \beta^*|0\rangle + \alpha^*|1\rangle$$

Qbit norm

For a given qbit, one require that its norm is 1. Meaning given $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ one require that its inner product with itself equals 1

$$\langle\psi|\psi\rangle = \begin{pmatrix} \alpha^* & \beta^* \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = |\alpha|^2 + |\beta|^2 = 1$$

Measurement Postulate

Given any physical measurements that is possible doing, the question that we are asking is what can we compute?

First to do so, we need to define what is a **measurement**. A **measurement** is an operator O which goes from a given Hilbert space into the same Hilbert space. For a given space $\mathbb{C}^n$ an operator on a this space is a matrix inside $\mathbb{C}^{n\times n}$.

Remark

We often like to put a hat above operator in order to have a difference between a scalar and an operator.

The question now is what is the outcome of our given measurement? The answer to this is the eigenvalues of our given operator.

Example Let us have a quantum state which is represent as an arrow, and we want to measure the z-projection of our quantum state. To do so let us create the operator

$$\hat{O} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and then apply it to our $|0\rangle$ state

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = (+1)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = +1|0\rangle$$

As we can see, $|0\rangle$ is an eigenvector with the eigenvalue 1.

By doing the same thing for $|1\rangle$ we can also see that it is also an eigenvector with it's eigenvalue -1.

1.1.1 Postulate (Measurement)

Let A be a Hermitian operator on a Hilbert space $\mathcal{H}$ with non-degenerate spectrum, so that $A|\lambda_i\rangle = \lambda_i|\lambda_i\rangle$ for an orthonormal basis $\{|\lambda_i\rangle\}$. Let the system be in a normalized state $|\varphi\rangle \in \mathcal{H}$. Then

1. The only possible outcomes of measuring A are the eigenvalues λ_i, and these depend on A alone
2. The outcome λ_i occurs with probability $p(\lambda_i) = |\langle\lambda_i|\varphi\rangle|^2$
3. Immediately after an outcome λ_i is obtained, the state of the system is $|\lambda_i\rangle$

The probability of outcome $|x\rangle$ and it's respective eigenvalue λ_x is given by

$$\text{Prob}(x) = |\langle x|\psi\rangle|^2$$

Let us start with $|\psi\rangle = |0\rangle$ We want to look for the probability of goin into each of our state $|0\rangle, |1\rangle$:

$$P(|0\rangle) = \langle 0|0\rangle = 1$$
$$P(|1\rangle) = \langle 1|1\rangle = 1$$

Now let us define another state given by

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

If we want to compute the probability to be $|1\rangle$ or $|0\rangle$ when measuring with our projector $\hat{Z}$:

$$P(|1\rangle) = |\langle\psi|1\rangle|^2 = \frac{1}{2}$$
$$P(|0\rangle) = |\langle\psi|0\rangle|^2 = \frac{1}{2}$$

As we can see, in order to predict what our initial quantum state is, one would need to have many equal state in order to create a prediction of the probability distribution.
A note that is important here is that by '*coincidence*' we have that the eigenvector of our $\hat{Z}$ are $|0\rangle$ and $|1\rangle$. What we can do to define our basis vector based on the eigenvector of our operator, which means that we don't have to define the eigenvector using numerically. To do so we can use the spectral theorem and state:

$$\hat{Z} = (+1)|0\rangle\langle 0| + (-1)|1\rangle\langle 1|$$

If your initial state is not a linear combination of the eigenvector of the operator, the state after measuring it using the operator will be updated with probabilities computed as above. Hence we can define our operator as if $|\psi\rangle$ was one of its eigenvector. $|\psi\rangle$ also has a common notation:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = |+\rangle$$

With also it's correspondent eigenvector

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

Then define $X = (+1)|+\rangle\langle+| + (-1)|-\rangle\langle-|$ when measuring X we are asking the question on the probability of the projection into those diagonal vector.

An interesting question now is starting from $|\varphi\rangle = |0\rangle$ we want to study the question of measuring X. First of all, because we are asking about X we know that the outcome state of our measurements will be either $|+\rangle$ or $|-\rangle$ with probabilities

$$P(|+\rangle) = \langle 0|+\rangle^2 = \frac{1}{2}$$
$$P(|-\rangle) = \langle 0|+\rangle^2 = \frac{1}{2}$$